

FitBuddy

AI Fitness Plan Generator using Gemini Models

2. Requirement Analysis Phase

Requirement Analysis is an important phase in the development of FitBuddy. In this phase, the requirements of the application are identified, studied, and documented before development begins.

Purpose of Requirement Analysis

- Understand what the FitBuddy application should do.
- Identify the needs of the users.
- Define the features and functions of the system.
- Identify the software and hardware requirements.
- Determine the expected input and output of the application.
- Reduce errors and confusion during later development stages.

Project Requirement Overview

FitBuddy is designed to accept basic fitness-related information from users and use Gemini Models to generate a personalized fitness plan. The requirements are divided into functional requirements, non-functional requirements, hardware requirements, software requirements, and user requirements.

Functional Requirements

Functional requirements describe the main activities that the FitBuddy system must perform.

- User Input: The system should allow users to enter relevant fitness information.
- Profile Details: The system may collect details such as age, fitness level, goals, available time, and workout preferences.
- Input Validation: The application should check whether required information has been entered correctly.
- AI Processing: The system should send suitable information or prompts to Gemini Models.
- Plan Generation: Gemini should generate a personalized fitness plan based on the provided requirements.
- Plan Display: The generated plan should be displayed in a simple and readable format.
- Error Handling: The application should show suitable messages when an error occurs.
- Plan Update: The system can support generating a new plan when the user's requirements change.

Expected Input

- Basic user information.
- Fitness level or experience.
- Fitness goal.
- Available workout time.
- Preferred workout activities or limitations.

Non-Functional Requirements

Non-functional requirements describe the quality, performance, usability, and security expected from the FitBuddy system.

1. Usability

- The interface should be simple and easy to understand.
- Users should be able to enter information without difficulty.
- The generated plan should be clearly organized.

2. Performance

- The application should respond within a reasonable time.
- The system should process user requests efficiently.

3. Reliability

- The application should work consistently for valid inputs.
- Errors should be handled without crashing the application.

4. Security and Privacy

- User-provided information should be handled carefully.
- Only necessary information should be collected.
- Sensitive information should not be unnecessarily exposed.

5. Maintainability

- The project should be easy to modify and improve.
- New AI features can be added in future versions.

Hardware and Software Requirements

Hardware Requirements

- Computer, laptop, or compatible mobile device.
- Sufficient RAM and storage to run the application.
- Internet connection for accessing Gemini Models when required.

Software Requirements

- Operating system such as Windows, Linux, macOS, Android, or another supported platform.
- Programming language and development environment suitable for the project.
- Gemini API or supported Gemini Model integration.
- Required libraries, frameworks, and development tools.
- Web browser if FitBuddy is implemented as a web application.

User Requirements

- The user should be able to provide basic fitness preferences.
- The user should understand the generated plan before following it.
- The application should provide clear information rather than complicated technical details.

System Input and Output

Input: User fitness details and preferences.

Processing: FitBuddy sends the relevant information to Gemini Models and processes the generated response.

Output: A structured and personalized fitness plan is displayed to the user.

Requirement Analysis Summary

The requirement analysis phase provides a clear foundation for developing FitBuddy. It identifies what the system should do, what resources are needed, and what users expect from the application.

Key Requirements

- Simple user-friendly interface.
- Collection of relevant fitness information.
- Gemini Model integration.
- Personalized fitness plan generation.
- Clear presentation of generated results.
- Input validation and error handling.
- Reasonable performance and reliability.
- Basic security and privacy considerations.

Basic Requirement Flow

User Details → Input Validation → FitBuddy Application → Gemini Model → AI Processing → Personalized Fitness Plan → Display to User

Expected Outcome

After completing the Requirement Analysis Phase, the development team will have a clear understanding of the functions, resources, and quality requirements of FitBuddy. These requirements can then be used as the foundation for the Project Design and Development phases.

Conclusion

Requirement Analysis is essential for building an effective FitBuddy application. Proper identification and documentation of requirements help the team develop a system that is useful, understandable, reliable, and suitable for generating personalized fitness plans using Gemini Models.

Project: FitBuddy – AI Fitness Plan Generator using Gemini Models